

KRISTOFFER DAVE A. TABONG

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kristoffer.tabong@kdtdatasolution.com | Data Scientist · Software Development Lead · Structural Engineer

PROFESSIONAL SUMMARY

Data Scientist and software development leader with twelve years of professional experience, currently leading the end-to-end delivery of custom software, data science, and AI solutions across web, mobile, and desktop platforms. Holds a Master of Science in Data Science and applies machine learning, analytics, and AI to build decision-driven products for clients across multiple industries. Began his career as a structural engineer at multinational firms, contributing to vertical structures and major infrastructure across Southeast Asia, North America, the Middle East, and Europe. Now President of a technology consultancy, an Associate Member of the Association of Structural Engineers of the Philippines, and a university lecturer in data science and analytics.

CORE COMPETENCIES & TECHNOLOGY STACK

Data Science & AI: Exploratory Data Analysis, Machine Learning, Deep Learning, Statistical Analysis, Natural Language Processing, Big Data & Cloud Computing

Software Development: Web (PHP / Laravel, React, Next.js, Blazor) · Mobile (React Native) · Desktop (C# / .NET) · Data & AI (Python, Django)

Data & Cloud: MySQL, AWS; integrations across Shopify, NFC / QR, and Facebook

Project Management: Waterfall & Agile delivery, time & budget management, technical writing

Structural Engineering: Vertical structures (residential, commercial, mixed-use); reinforced concrete & structural steel design; schematic design to construction documentation

SOFTWARE DEVELOPMENT & DATA SCIENCE PROJECTS

Data Science & AI

Education LMS with AI

AI & Education

A learning-management platform for higher education with AI built in — supporting class modules, outcome and curriculum mapping, Bloom's-level alignment checks, and automated quality-improvement reporting for faculty and program heads.

Stack: Python + Django · Integrated AI

AI Chatbot Platform

AI & Automation

An AI chatbot that businesses train on their own knowledge base, then deploy across their website and Facebook Messenger — answering customers in natural, conversational language, including Taglish.

Stack: React · React Native · MySQL · AWS · Facebook

TABS — Structural Analysis & Design

In-House Product (Ongoing)

In-house structural engineering software — a 3D modeling environment for framing, loads, and supports with built-in analysis, design, and reporting, paired with local and cloud AI engines.

Stack: Electron · C# / C++ · Python · Local & Cloud AI

Insurance Portfolio Management

FinTech (Ongoing)

A white-labeled treasury and risk platform for an insurance client — bringing cash position, investments, receivables, credit & collection, and reinsurance settlement into a single command center.

Stack: React · Next.js · MySQL

Web & Mobile Platforms

ConstructPro — Project Management Platform

Web & Mobile

A web and mobile suite for planning projects, tracking tasks and schedules, managing invoices and expenses, and monitoring budgets against actuals — giving managers a single view of every project's health.

Stack: PHP / Laravel · React Native · MySQL · AWS

UAP National Portal (UAP Web 2.0)

Web & Mobile

A membership portal for the United Architects of the Philippines — handling registration and records, chapter management, certificate issuance, events, news, e-library, and a designer's directory across web and mobile, serving tens of thousands of members.

Stack: PHP / Laravel · React Native · MySQL · AWS

DMPR — Rapid Structural Assessment

Disaster Response

A field tool for the Association of Structural Engineers of the Philippines, recording pre- and post-earthquake evaluations of buildings on mobile to rapidly triage safety, feeding a dashboard that maps inspections, postings, and damage by incident.

Stack: PHP / Laravel · React Native · MySQL · AWS

Warehouse Management System

Web & Mobile

An inventory and fulfillment system with QR-sticker tagging, so every item and unit can be scanned in and out — covering catalog, stock levels, slotting, and inbound/outbound flows across multiple warehouses.

Stack: React · React Native · MySQL · AWS · QR

Digital Card System

Connected Experience

A digital networking-card app paired with physical NFC cards — a single tap shares a fully branded profile, replacing paper business cards.

Stack: React · React Native · MySQL · AWS · NFC

Industrial & E-Commerce

Pipe Manufacturing Software

Industrial Software

Desktop engineering software that configures pipe-extrusion programs — profiles, materials, tooling, and layer parameters — and computes production time and machine settings, paired with a Blazor web layer.

Stack: C# / .NET (WinForms) · Blazor

E-Commerce Platform

E-Commerce

A Shopify storefront with custom integrations for a phone and accessory retailer — organized by device brand, with product catalog, promotions, and a streamlined checkout.

Stack: Shopify · Custom Integrations

PROFESSIONAL EXPERIENCE

KDT Network and Data Solution

January 2024 – Present

President · Quezon City, PH

- **Lead overall operations** of the company and the end-to-end delivery of custom software, data science, and AI solutions — from scope to launch to support on AWS.
- **Direct** an engineering-minded team across four service pillars: software development, data science & AI, architecture & engineering, and technical training.
- **Provide** technical consultancy to SMEs, organizations, and stakeholders across IT, data science, civil engineering, and construction.
- **Oversee** NPC-registered, privacy-by-design practices (DPO / DPS compliant) across every client engagement.

University of the Philippines – OVPD-ODPI

January 2026 – Present

Junior Managing Engineer (Structural) - Consultant

- **Manage** complex structural projects in alignment with university development goals, overseeing deliverables and the preparation and review of detailed structural engineering drawings.
- **Review, approve,** and sign structural and civil engineering plans submitted by UP's constituent universities for consistency with the Master Development Plan, design guidelines, and policies.
- **Develop** strategies to improve structural and civil engineering practices within the office.

Multi Development and Construction Corporation

June 2021 – December 2023

Construction Systems Analytics Director · Taguig City, PH

- **Directed** the organization's construction data-transformation operations, owning the full technology stack — process and equipment design, software, hardware, data, and cloud infrastructure.
- **Built** management dashboards and data presentations to communicate the technology vision to customer stakeholders.
- **Led** long-term capability planning, recruitment, and organizational development to enhance project management through data and technology.

University of Santo Tomas – Dept. of Math and Physics

August 2022 – June 2025

Adjunct Professor · Manila, PH

- **Taught** Data Science and Analytics, Exploratory Data Analysis, Artificial Intelligence, Information Visualization, and Computer-Aided Statistical Inference.
- **Developed** course materials and delivered instruction aligned with university standards and regulations.

STRUCTURAL ENGINEERING CAREER (SUMMARY)

Began his civil and structural engineering career in 2014, progressing from graduate engineer to structural design engineer at multinational firms. Performed finite-element analysis, reinforced concrete and structural steel design, and full schematic-to-construction documentation for vertical structures and major infrastructure across Southeast Asia, North America, the Middle East, and Europe.

- **KTP Philippines ROHQ** (a member of Surbana Jurong) — Structural Engineer · Mandaluyong, PH · Nov 2018 – Mar 2020
- **Arcadis GEC Manila** — Structural Design Engineer · Makati, PH · Dec 2017 – Nov 2018
- **Arcadis GEC Manila** — Assistant Structural Design Engineer · Makati, PH · Aug 2016 – Dec 2017
- **Fluor Daniel Inc.** — Associate Design Engineer I · Muntinlupa, PH · Jul 2015 – Aug 2016
- **Arcadis** (formerly Hyder Consulting) — Graduate Structural Design Engineer · Netherlands · Jan 2014 – Jul 2015

Notable Structural Projects

- Philippine Institute for Development Studies (10-story with basement) and a 10-story University Residence, Diliman, Quezon City; UP Tacloban Academic Building and Dormitory; Moriones Head Monument, Marinduque.
- Surbana Jurong Campus, Singapore; Doha Metro Red Line; Security Forces Medical Center, Riyadh, KSA; Children's Museum, Qatar; SASOL LCCP Gas-to-Liquid & Ethane Cracker, Louisiana, USA; Saudi British Bank Building, Riyadh; Al Falah Mall, Abu Dhabi.
- Kainomayan Public Hospital and Botolan Municipal Hall, Zambales; DOTr Two-Story Office Building, Micronesia; Monolithic Dome Warehouse, Tuguegarao, Cagayan; and over 30 residential projects.

EDUCATION

Asian Institute of Management – Makati City, PH

Master of Science in Data Science | March 2020 – June 2021

University of Santo Tomas – Manila, PH

Bachelor of Science in Civil Engineering, Magna Cum Laude | March 2008 – June 2013

Massachusetts Institute of Technology – MIT xPro (online), United States

Architecture and Systems Engineering: Models and Methods to Manage Complex Systems | Sept 2021 – March 2022

CERTIFICATIONS

- **AWS Academy** AWS Academy Graduate – Cloud Architecting · May 2022
- **NCEES** — Passed PE Civil, Structural Engineering Exam, California, USA · April 2018
- **NCEES** — Passed FE Civil Engineering Exam, California, USA · September 2017
- **DPWH** — Materials Engineer I (Accredited Contractors and Consultant Materials Engineer) · May 2015
- **PRC** — 8th Placer, Master Plumber Board Examination · September 2014
- **PRC** — 3rd Placer, Civil Engineering Board Examination · November 2013

ORGANIZATIONS

- **Association of Structural Engineers of the Philippines (ASEP)** — Associate Member · September 2024 – Present
- **Rotary Club of the Philippines, Commonwealth** — Member · June 2025 – Present